

The Role of Machine Learning in Predicting and Understanding Inflation Dynamics: Insights from the COVID-19 Pandemic

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Abstract—Inflation, or the increase in prices of goods and services, has become a significant issue during the Covid-19 pandemic. I explore diverse Machine Learning techniques to forecast inflation in the USA, prioritizing understanding over mere replication, dissecting method mechanics and key factors during periods of rapid price escalation. The findings indicate that ML models are indeed helpful in predicting inflation in the USA. Specifically, the more advanced non-linear models like random forest and ada boost perform better, especially during periods of high inflation. During high inflation, traditional factors like price trends and the Phillips curve still have explanatory power, but inflation itself tends to drive prices more prominently due to the unique dynamics of such periods.

Index Terms—Ada Boost, Inflation, Machine Learning, Phillips Curve, Random Forest

I. INTRODUCTION

Inflation, a rise in prices, has become a significant concern worldwide following the pandemic. It has affected both advanced and emerging economies, prompting a keen focus on monetary policy. Accurately predicting inflation in the short, medium, and long term is crucial for effective monetary policy implementation [1]–[3]. High inflation erodes purchasing power, dampens economic growth, and breeds uncertainty, while low inflation can hinder consumption and investment, impeding growth and job creation. Japan's prolonged economic stagnation due to deflation serves as a poignant example. The USA has seen varying inflation rates over the last decade, from lows after the financial crisis to highs post-Covid-19, necessitating accurate forecasts by central banks to adjust policies for economic stability.

Therefore, in this study, I examine various Machine Learning (ML) [4]–[8] models and a diverse set of variables to forecast inflation in the USA. I assess various machine learning models' predictive performance using a diverse dataset containing economic, financial, and market indicators. The analysis spans nearly 44 years, from February 1968 to December 2011, for model estimation. Subsequently, the models are tested over a ten-year period from January 2012 to July 2022, segmented into various inflation regimes. I employ the "Shapley Values" technique to assess variable importance and delve into model performance across different inflation

scenarios. Finally, I discuss the limitations of the study, avenues for future research, and draw conclusions based on the findings.

The paper outlines related studies, conducts an empirical study using datasets, offers a technique with metrics evaluation, analyzes results, presents limitations, and concludes with ideas.

II. RELATED WORKS

The recent surge in inflation rates globally has caught many analysts and international organizations off guard after years of relatively low prices [9]–[11]. This unexpected inflationary trend, particularly during the post-pandemic recovery, has raised significant concerns. While pinpointing a single cause remains elusive, multiple factors have been identified simultaneously. [12] provide insights into some of the prominent contributors to this inflationary episode. Some studies [13] suggest that the quicker-than-expected economic rebound has been a primary driver of inflation. Heightened uncertainty during the pandemic led to substantial policy interventions, bolstering fiscal support. Fiscal policies spurred demand post-Covid, but global supply chain disruptions intensified inflation amid sectorial shifts. Some sectors faced restrictions while others experienced surges in demand (e.g., teleworking, food), impacting relative prices. Persistent low unemployment amplifies demand pressures, adhering to the traditional Phillips curve theory, where tightening labor markets drive up wages and subsequently elevate production costs and consumer prices.

Some studies, such as [14], posit that shifts are occurring in how consumers and firms form price expectations. While some held high expectations, they were disregarded due to central bank overconfidence following years of meeting targets. Preceding the Covid-19 pandemic, prolonged loose monetary policies and expansive fiscal measures likely contributed to the resurgence of inflation. Therefore, forecasting inflation remains challenging, with varying success among different models and indicators. While some studies, like [15], have identified potential predictors of USA inflation among demand indicators, others, such as [16], have found limitations in traditional models like the Phillips curve. These divergent findings have spurred investigations into alternative models and variables. While some studies explore

alternative variables to elucidate inflation, others delve into the predictive capabilities of existing models. [17] note the examination of expectation variables, commodity prices, financial indicators, and supply-side factors as potential avenues for understanding inflation dynamics.

Advancements in technology and ML methods have fueled significant growth in predicting inflation by enabling the accumulation of larger datasets, enhanced processing capabilities, and ongoing algorithm development. In the realm of using ML for inflation forecasting, there exist various examples. For instance, [18] applied different ML techniques, such as LASSO, Ada-LASSO, and random forest, to forecast inflation in Brazil. Similarly, [17] focused on the USA, utilizing a sizable database divided into eight groups. Although similar to the database used in this study, I have updated mine until 2022 to enable testing in recent high inflation scenarios, particularly from mid-2020 to 2022, following the acceleration of inflation post-Covid-19. A more recent study by [19] concentrated on UK inflation, employing a methodology akin to mine to assess the impact of different variables (known as "Shapley Values"). This study employs accurate ML models to assess variable contributions in explaining inflation, comparing findings between the US and UK, contributing to the ML-driven inflation forecasting literature by analyzing recent inflationary periods and variable relevance during normal and high inflation episodes.

III. EMPIRICAL STUDY

In this section, I outline the dataset used, describe the different models I will consider, and explain how I will assess the accuracy of the models' predictions and the importance of the variables under different conditions. The primary source of data is the Federal Reserve Economic Data (FRED) database, created and managed by the US Federal Reserve, as referenced in [17]. After cleaning and transforming the data, I employ ML models to predict inflation at various monthly intervals within the training sample using Python and the scikit-learn package. I evaluate forecasting accuracy using RMSE and MAE, comparing against a Naive model across various test sets representing total, low, and high inflation scenarios. Furthermore, I evaluate the significance of the variables in the best models through the utilization of "Shapley Values", examining their relevance across different testing samples.

A. Data Analysis

The primary data source utilized in this study comprises monthly indicators pertaining to the USA, acquired from the FRED dataset. The Federal Reserve Bank of St. Louis research division oversees a vast online repository housing hundreds of thousands of economic data time series from diverse national and international sources.

The Dependent Variable: Inflation under Different Regimes

The primary focus of this study revolves around analyzing the USA inflation rate, with particular emphasis on the consumer price index for all urban consumers (all items), a metric compiled and adjusted seasonally by the Bureau of Labor

Statistics. The examination focuses on monthly rates of the adjusted series to smooth out annual statistical fluctuations, spanning February 1968 to July 2022, with an average monthly rate of 0.33% (equivalent to 4.1% annually) over 654 observations. Table I: Dataset segmented into distinct samples and periods:

- **Training Sample (1968-2011):** Between February 1968 and December 2011, constituting 81% of the data, the average monthly inflation rate was 0.36%, exceeding the overall average, with an annual rate of 4.5%.
- **Testing Sample (2012-2022):** The testing sample from January 2012 to July 2022 shows an average monthly inflation rate of 0.2% and an annual rate of 2.5%, closely aligning with the USA Federal Reserve's 2.0% target, with two distinct periods identified.
- **Low Inflation Test Sample (2012-2019):** From January 2012 to January 2020, the aftermath of the 2009 financial crisis and economic conditions yielded a monthly inflation average of 0.14% and an annual average of 1.60%, marked by occasional episodes of deflation.
- **High Inflation Test Sample Period (July 2020-2022):** From July 2020 to July 2022, economic repercussions of the Covid-19 pandemic led to a significant surge in inflation, averaging 0.55% monthly and 5.40% annually, surpassing the Federal Reserve's target.

Various inflationary regimes enable a thorough assessment of machine learning models and explanatory variables' predictive power across six different horizons. This methodological approach aligns with prior research by [18] and [17], with a specific focus on evaluating model accuracy and variable performance during recent periods of high inflation, constrained by data availability limitations.

TABLE I
DATA INFORMATION

Data	Dates	Monthly Inflation	Yearly Inflation
Total Sample	Jan 1968 - July 2022	0.33%	4.10%
Training Sample	Jan 1968 - Dec 2011	0.36%	4.50%
Test Sample	Jan 2012 - July 2022	0.20%	2.50%
Low Inflation Test Sample	Jan 2012 - July 2020	0.14%	1.60%
High Inflation Test Sample	June 2020 - July 2022	0.55%	5.40%

1) The Explanatory Variable: Different Groups

The FRED dataset contains a large number of different variables collected over a long period. The dataset we're using in this study was proposed by [17] and includes 99 variables grouped into various categories, including one main variable we're interested in and eight groups of other variables. To

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make analysis easier, we've transformed all the data to make sure it's consistent over time. Different groups of variables are:

- **Price components:** Include vital consumer price indexes for different goods/services. We use these to understand the importance of groups of products and to see how prices change over time.
- **Price costs:** This includes prices that companies pay for goods and materials, like oil prices. It helps us see how companies pass on their costs to consumers.
- **Unemployment or Labor conditions:** Understanding the job market informs central banks like the Federal Reserve about potential wage pressures and aligns with the Phillips Curve theory, incorporating industry-specific wage data.
- **Activity Indicators:** These are economic indicators that gauge overall economic activity, helping assess inflation risks by measuring production levels in factories, contrasting with labor conditions indicators.
- **Labor Force:** Similar to activity indicators and reflects position in the economic cycle. Includes information on how many people are employed in different industries and how much they're paid.
- **Money and credit:** Some economists think that inflation, or rising prices, is caused by changes in economic money. Includes variables to test that idea.
- **Interest rates:** This group includes different kinds of interest rates, like the government pays on its debt and companies pay to debt. Interest rates are also an important tool for central banks to control inflation.
- **Housing market:** Changes in the housing market, such as falling house prices during the financial crisis, can impact inflation dynamics.

IV. METHODOLOGY: GENERAL EXPRESSION

Consider the model for monthly inflation at different time horizons (h), say 1, 2, 3, 4, 5, and 6 months. Therefore, in this study, the inflation in a month $t + h$ (where h represents the horizon) is denoted by π_{t+h} , and x_t is an n -dimensional vector comprising explanatory factors and/or common factors. Similarly, let G_h signifies various models linking the explanatory factors at time t to the future inflation at time $t + h$. Additionally, u_{t+h} denotes a random error with zero mean. The function G_h could be a single model or a combination of different specifications as shown in Equation (1).

$$\pi_{t+h} = G_h(x_t) + u_{t+h} \quad h = 1, \dots, H; t = 1, 2, \dots, T \quad (1)$$

A general formula encapsulating the essence of most algorithms can be outlined as described by [20]. Essentially, it comprises a function class $f(x)$, which might entail a linear function, a factor, a tree, or a neural network, among others. Additionally, there's a component called the "regularizer" or $R(f)$, which dictates how complexity—defined as the presence of numerous coefficients and variables in the model—is handled. As per [20], selecting the prediction function involves two steps: first, choose the best in-sample loss-minimizing function given a certain complexity level, and secondly, estimating the optimal complexity level through empirical tuning. In the general formula provided, we adopt

the deviance between the data and the model fit (denoted as Loss function), and we assume a generic form for the regularizer, denoted as $\lambda \|f\|_{l_q}$. The "regularizer" can be represented as the product of a regularization strength parameter (λ) and its functional shape. It's worth noting that by introducing a penalty on the magnitude of each coefficient (β_k), this approach penalizes complexity since these coefficients govern the impact of input variables (X_i) on predicting the outcome (y_i). We can frame the problem in the general forms of Equations (2) and (3), which can be tackled using numerical optimization techniques such as Newton's method or Gradient Descent.

$$\text{minimize } \sum_{i=1}^n L(y_{i,t+h}, f(X_{i,t})) + \lambda \|f\|_{l_q} \quad (2)$$

$$\beta = \text{argmin} \{ \sum_{i=1}^n 1(y_{i,t+h} - \beta X_{i,t})^2 + \lambda \sum_k c(\beta_k) \} \quad (3)$$

A. Evaluation Metrics

After estimating models, evaluating their significance entails comparing testing errors to actual data and assessing variable importance using Shapley values, which measure each variable's average contribution to the final forecast.

1) Evaluation of Models: Comparing Errors

Comparison of predicted values from different models with actual data in test samples helps determine the accuracy of each model across various timeframes. The inflation forecast for a given month by a model with information available up to time t is represented by $\hat{\pi}_{t+h/t} = \hat{G}_{h,t}(x_t)$. We evaluate the models using two common metrics: RMSE and MAE. Including both MAE and RMSE in our analysis is crucial to ensure that our findings are robust and not heavily influenced by a few large forecasting discrepancies. Additionally, we compare the performance of each inflation model across different horizons with that of a benchmark model, specifically a naive model depicted in Equation (4). This model solely relies on the most recent monthly inflation rate to predict future inflation at various horizons.

$$\hat{\pi}_{t+h/t} = \pi_t \quad (4)$$

Using the naive model as a benchmark serves to establish a minimum standard for the effectiveness of alternative models. Essentially, any model we propose should outperform this baseline model at every horizon. To quantify this comparison, consistent with existing literature, we compute the ratios of specific model performance relative to that of the Naive model as shown in Equation (5) and (6).

$$\{\text{RatioRMSE}\} = \frac{RMSE_{m,h}}{RMSE_{Naive,h}} \quad (5)$$

$$\{\text{RatioMAE}\} = \frac{MAE_{m,h}}{MAE_{Naive,h}} \quad (6)$$

2) Evaluation of Variables: Analysis of Shapley Values

An essential consideration in the realm of ML models is evaluating the relevance of variables. This involves understanding the significance of each variable for the model's performance and its impact on generating forecasts. While assessing the importance of explanatory variables in a straightforward linear model is relatively simple, untangling

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their influence in more complex non-linear models like random forests or neural networks poses a challenge. A recent tool known as "SHAP" has emerged as an effective approach for discerning the importance of variables within a model, as outlined by [19]. SHAP is grounded in Shapley values, derived from a mathematical theory ensuring equitable distribution of variable contributions. By decomposing a particular prediction into Shapley values, this method offers local interpretability, shedding light on the individual roles of variables within intricate models. Essentially, the method dissects predictions generated by any algorithm into variable contributions, revealing the Shapley values of each, thus breaking down individual predictions of observation into the contributions of the individual features or variables. Equation (7) illustrates the decomposition of predictions into variable contributions, underscoring the method's capability to analyze individual variables' impacts on model predictions.

$$f(x_t) = \sum_{k=0}^N \phi_k(x_t) \quad (7)$$

V. RESULT ANALYSIS

In this section, the primary findings of various algorithms are elucidated and juxtaposed with those of the benchmark model. Commencing with the most straightforward approach, the outcomes of the Naive model across different timeframes—namely, one, three, and six months—are delineated in Fig. 1. Subsequent algorithms are contextualized in relation to this baseline, enabling a comparative analysis. The results presented in Table II unveil that the RMSE for the overall inflation dataset exhibits relative consistency and a moderate elevation from 0.34% to 0.35% when transitioning from shorter (one month) to medium-term (six months) perspectives. Notably, given the average monthly inflation rate of the test sample stands at 0.20%, the recorded error registers as relatively elevated. Upon disaggregating the error term across sub-samples characterized by varying inflation levels—low and high—a nuanced pattern emerges. While the error remains stable across both regimes, it manifests distinct magnitudes, hovering around 0.28%-0.31% during low inflation periods and escalating notably to 0.43%-0.45% amidst high inflation epochs. In essence, the discerned errors underscore the formidable challenge of forecasting inflation via simplistic Naive models, with high inflation periods posing heightened difficulty, thus highlighting the potential for enhancement through ML paradigms.

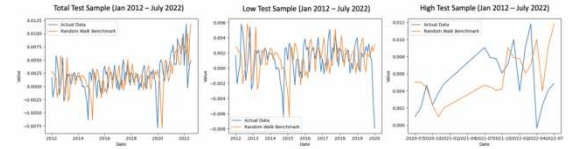


Fig. 1. Naive benchmark model for each of the three test samples

TABLE II
NAIVE BENCHMARK MODEL RMSE STATISTICS IN 1, 3 & 6 MONTHS

Benchmark Base	1 Month	3 Months	6 Months
Total	0.34%	0.35%	0.35%
Low	0.28%	0.28%	0.31%
High	0.43%	0.41%	0.45%

Moving forward, the efficacy of alternative ML models at the short-term horizon (one month) is depicted in Fig. 2 and Table III. Herein, the RMSE ratios relative to the benchmark model—Naive—are meticulously scrutinized. Notably, a salient observation pertains to the pronounced elevation in RMSE Ratio across the total sample in the case of linear models, primarily attributed to the inclusion of severe Covid-19 months within the inflation dataset, thus accentuating the traditional linear models' struggle in capturing or elucidating significant price fluctuations. This disparity is also evident among some regularized models, albeit to a lesser extent. Conversely, non-linear models such as random forest and AdaBoost exhibit superior performance, manifesting lower RMSE errors. Importantly, upon segmenting the analysis into low and high inflation periods, distinct trends emerge. Specifically, during low inflation periods, linear and regularized models demonstrate comparable or marginally superior performance relative to non-linear counterparts. However, amidst high inflation phases, non-linear models showcase marked superiority, underscoring their efficacy particularly in the short run (one month).

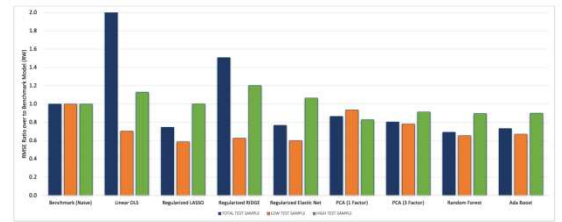


Fig. 2. RMSE ratio to benchmark model during a period of one month

TABLE III
RMSE RATIOS TO BENCHMARK MODEL IN 1 MONTH HORIZON

RMSE 1 Month	Benchmark k	Linear Ols	Regularise d Lasso	Regularise d Rigid	Regularised Elastic Net	PCA (1)	PCA (3)	Random Forest	Boost
Total Test Sample	1	8.53	0.75	1.51	0.77	0.86	0.80	0.69	0.73
Lowtest Sample	1	0.70	0.59	0.63	0.60	0.93	0.78	0.66	0.67
High Test Sample	1	1.13	1.00	1.20	1.06	0.83	0.91	0.90	0.90

In Fig. 3 and Table IV, we present the RMSE ratio compared to a benchmark model over a short to medium timeframe (3 months). Generally, the RMSE ratios align with those observed in the short-term (1 month) analysis. Linear models encounter challenges, often failing to surpass the benchmark, especially in explaining significant price fluctuations. This

trend extends to the Ridge model. However, these models demonstrate relatively better performance in periods of low inflation. Notably, regularization models like LASSO and Elastic Net outshine linear models and perform comparably to nonlinear models during these periods. Nonlinear models exhibit superior predictive abilities during high inflation periods, with Principal Component Analysis (PCA) (1 factor)

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and AdaBoost emerging as top performers. Results are akin to those observed in very short-term analyses, albeit with narrower discrepancies between nonlinear and linear models in high inflation periods. Nevertheless, a slight advantage persists for nonlinear models in total and high inflation test samples. The comparative analysis results across all horizons, highlighting the frequency of model outperformance relative to the Naive Model. Most models surpass the naive approach in the full test sample, except for traditional linear models, owing to the complex Covid-19-induced environment. ML models prove valuable predictors during the full test sample, particularly in low inflation years, showcasing the benefits of ML adoption. A significant finding is the superiority of nonlinear models during high inflation periods. On average, top models like AdaBoost and random forest outperform the Naive model in 94% of cases, with even higher success rates in total and low inflation test sets. Notably, nonlinear models

consistently outperform the naive approach during high inflation periods, aligning with prior research highlighting random forest's efficacy in USA inflation forecasting. Furthermore, introducing regularization to linear models, notably Lasso and Elastic Net, yields gains, which is unsurprising given the high variable count in our analysis.

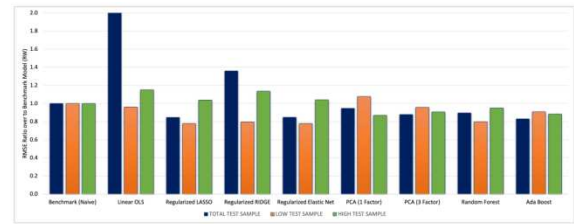


Fig. 3. RMSE ratio to benchmark model during a period of three month

TABLE IV
RMSE RATIOS TO BENCHMARK MODEL IN 3 MONTH HORIZON

RMSE 1 Month	Benchmark	Linear Ols	Regularised Lasso	Regularised Rigid	Regularised Elastic Net	PC A (1)	PC A (3)	Random Forest	Boost
Total Test Sample	1	11.59	0.35	1.36	0.85	0.95	0.80	0.88	0.83
Low Test Sample	1	0.96	0.78	0.80	0.78	0.78	0.78	0.96	0.91
High Test Sample	1	1.15	1.04	1.14	1.04	0.83	1.04	0.95	0.88

A. Relevance of Variables: The Shapely Values

This study delves into analyzing the significance of various factors in the most effective machine learning model for predicting inflation, focusing on primary and sets of factors within the random forest model, and employing Shapley values to evaluate their impact on the forecast. This study aims to ascertain if variable importance remains consistent across varying time periods and distinct inflationary contexts, enhancing comprehension of factors influencing inflation forecasting. Shapley values are computed for different test samples (total inflation, low inflation, high inflation) to evaluate variable importance, and their additivity allows for determining combined contributions of variable sets to the model's forecast. The random forest forecast for the 2012-2022 period generally reflects empirical and theoretical discussions, slightly overestimating the average inflation rate of 0.26% compared to the actual 0.20%. During this period, factors like price dynamics, labor market conditions, and indicators of economic activity, including unemployment, play crucial roles in shaping inflation, aligning with the traditional understanding of the Phillips curve's significance.

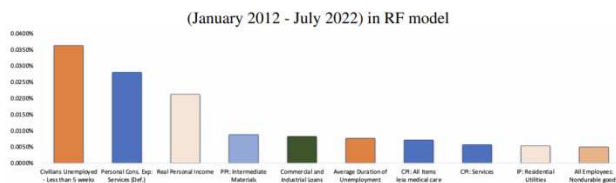


Fig. 4. Top 10 factors in the random forest model that have the most influence in the total inflation test set (January 2012 - July 2022)

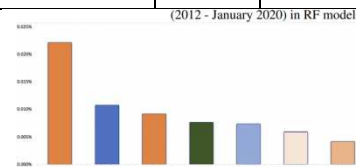


Fig. 5. Top 10 factors in the random forest model that have the most influence in the total inflation test set (January 2012 - January 2020) (July 2020 - July 2022) in RF model

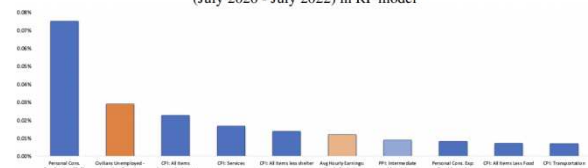


Fig. 6. Top 10 factors in the random forest model that have the most influence in the total inflation test set (July 2020 - July 2022)

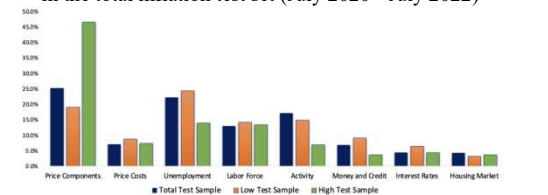


Fig. 7. Comparison of the three test sets' "total" contributions from various variable groups to inflation in the random forest model

VI. LIMITATIONS AND FUTURE WORKS

While the approach I adopt in this study offers several advantages, it is also subject to certain limitations. Initially, despite having a substantial and balanced dataset consistent with prevailing practices in ML research (81% for training and 19% for testing), the evaluation of the "high inflation" dataset is constrained by its reliance on only two years of data. Conversely, the assessment of the low inflation dataset spans over more than eight years. The scarcity of recent data underscores the need to delve into historical high inflation

periods to validate findings, while the diverse array of explanatory variables could affect the significance of Shapley values due to varying influences of specific factors. To address concerns and enhance the study, employ techniques like pre-estimation variable selection and dimensionality reduction (e.g., PCA), incorporating secondary benchmark models (e.g., rolling averages), alternative models like XGBoost or neural networks, and extending analysis to other countries like Spain despite data scarcity challenges [17].

VII. CONCLUSION

This study presents significant findings regarding the application of ML models in predicting crucial economic variables like inflation. Generally, these models exhibit the capacity to enhance inflation forecasts compared to simplistic approaches across various short to medium-term timeframes. Consequently, there is promise in leveraging ML for enhancing inflation predictions, not only in the USA but also in other countries and for diverse economic indicators. With the continuous advancements in ML and the availability of big data, the relevance of these models is poised to grow in the foreseeable future. Furthermore, the results highlight the superior performance of non-linear models like random forest and ada boost in forecasting inflation in the USA. These models consistently outperform the Naive model across different test sets, demonstrating lower RMSE and MAE errors. Notably, their superiority is more pronounced during periods of high inflation, potentially indicating the non-linear nature of recent inflation trends. Moreover, the analysis of variable importance in the best-performing model yields insightful results relevant to the ongoing discourse on the drivers of recent inflation. During high inflation periods, price dynamics gain prominence while indicators linked to the Phillips Curve hypothesis lose relevance, highlighting shifting variable importance in inflation forecasting, with monetary, financial, and housing market indicators showing minimal impact in recent years in the USA.

VIII. DECLARATIONS

- A. Funding:** No funds, grants, or other support was received.
- B. Conflict of Interest:** The authors declares no known competing interests.
- C. Data Availability:** Data will be made on prior request.
- D. Code Availability:** Code will be made on reasonable request.

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